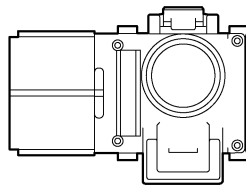
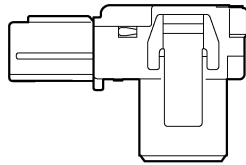


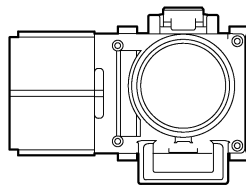
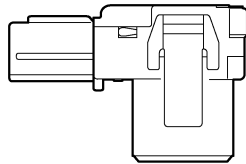
■ CONSTRUCTION AND OPERATION

1. Ultrasonic Sensor

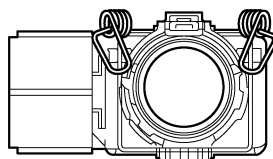
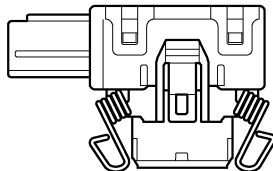
- The ultrasonic sensor consists of the microphone that transmits and receives ultrasonic wave and circuit portion.
- The circuit portion is filled with silicone or urethane to prevent water from entering.



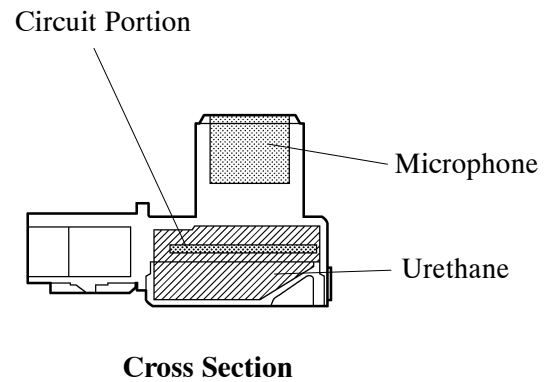
**Ultrasonic Sensor
(Front Corner LH and RH)**



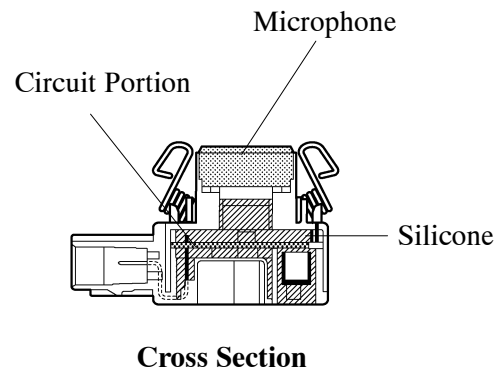
**Ultrasonic Sensor
(Rear Corner LH and RH)**



**Ultrasonic Sensor
(Rear Center LH and RH)**



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2. Clearance Warning Buzzer

- Depending on the detection distance and the detection area, the sound pattern of the clearance warning buzzer will vary.

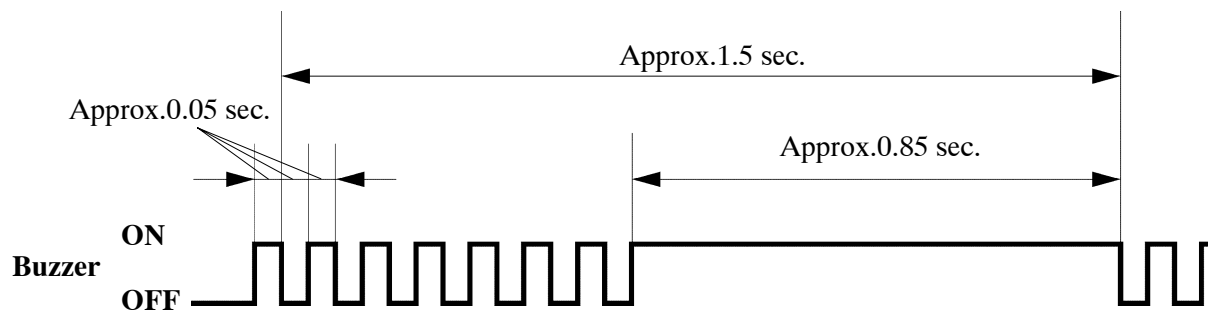
Detection Area	Detection Distance cm (in.)		Buzzer Sound Pattern ON/ OFF Time (msec.)
Front Corner	Long	60 ± 6 to 47.5 ± 5 (23.6 ± 2.4 to 18.7 ± 2)	150 / 150
	Middle	47.5 ± 5 to 35 ± 4 (18.7 ± 2 to 13.8 ± 1.6)	75 / 75
	Short	35 ± 4 or less (13.8 ± 1.6 or less)	Continuous Sound
Rear Corner	Long	60 ± 6 to 37.5 ± 4 (23.6 ± 2.4 to 14.8 ± 1.6)	150 / 150
	Middle	37.5 ± 4 to 30 ± 3 (14.8 ± 1.6 to 12 ± 1.2)	75 / 75
	Short	30 ± 3 or less (12 ± 1.2 or less)	Continuous Sound
Rear Center	Longest	150 ± 15 to 60 ± 6 (59 ± 5.9 to 23.6 ± 2.4)	150 / 650
	Long	60 ± 6 to 45 ± 5 (23.6 ± 2.4 to 17.7 ± 2)	150 / 150
	Middle	45 ± 5 to 35 ± 4 (17.7 ± 2 to 13.8 ± 1.6)	75 / 75
	Short	35 ± 4 or less (13.8 ± 1.6 or less)	Continuous Sound

- When the front and rear sensors detect obstacles at the same time, the buzzer sounds as follows.

ON/OFF Time (msec.)

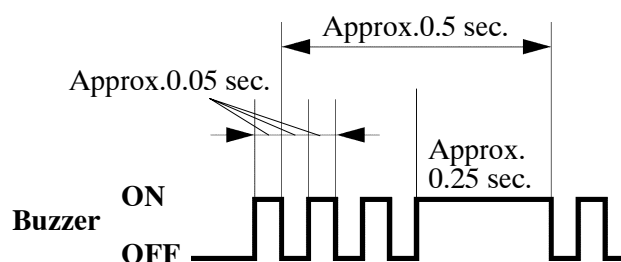
Front \ Rear	Short	Middle	Long	Not Detected
Short	Buzzer Timing 2	Buzzer Timing 1	Buzzer Timing 1	Continues Sound
Middle	Buzzer Timing 1	75/ 75	75/ 75	75/ 75
Long	Buzzer Timing 1	75/ 75	150/ 150	150/ 150
Longest	Buzzer Timing 1	75/ 75	150/ 150	400/ 400
Not Detected	Continues Sound	75/ 75	150/ 150	None

► Buzzer Timing 1 ◀



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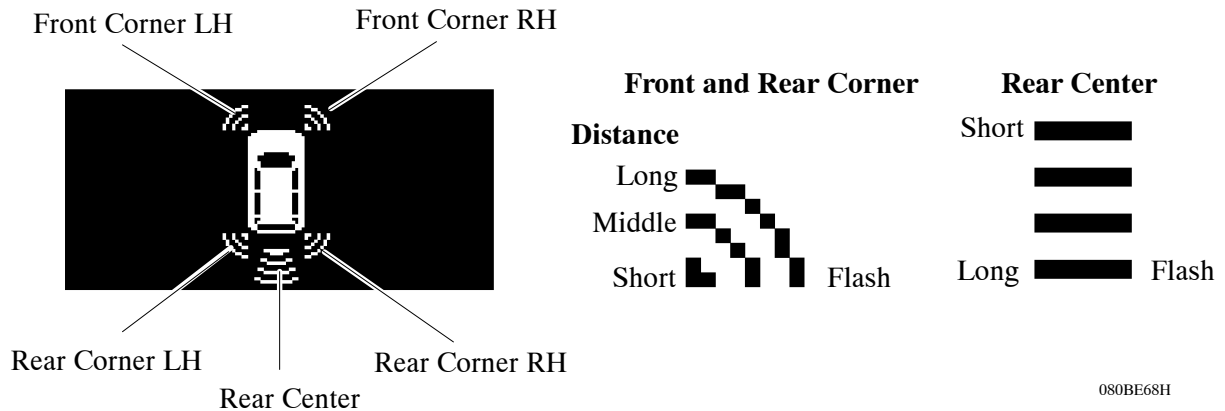
► Buzzer Timing 2 ◀



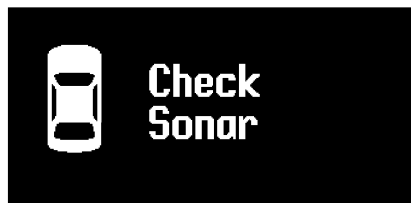
0140BE76C

3. Multi-information Display

- The items displayed by the multi-information display are the location of the obstacle, the approximate distance between the vehicle and the obstacle, and warning messages relating to sensor malfunction, sensor freezing, or presence of dirt on the sensor. When any warning messages are displayed, the master warning light illuminates and the clearance warning buzzer sounds.
- The number of lines shown on the display changes based on the actual distance and flashes when the distance is short.



► Warning Message ◀



Ultrasonic Sensor Malfunction Warning*



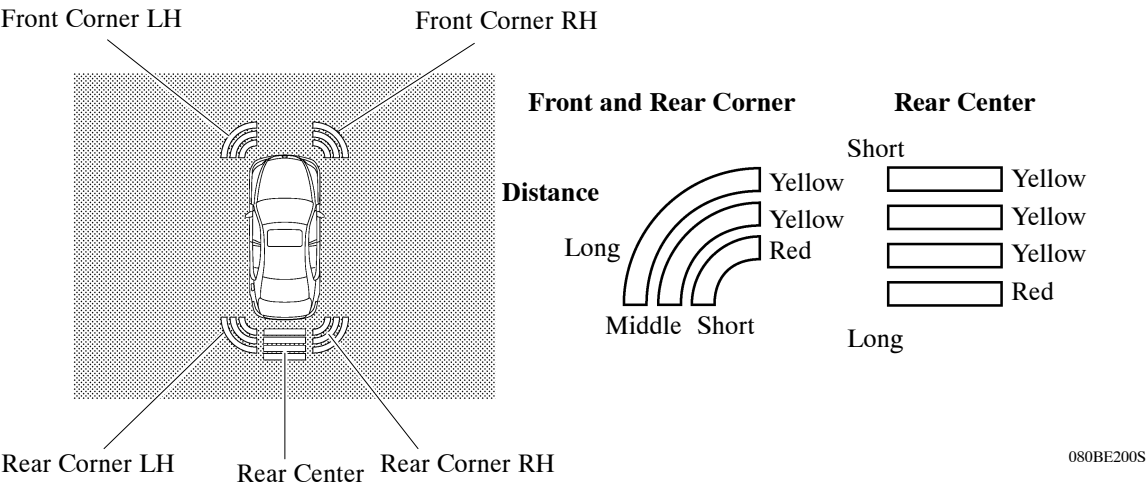
Ultrasonic Sensor Freeze/Contamination Warning*

080BE69S

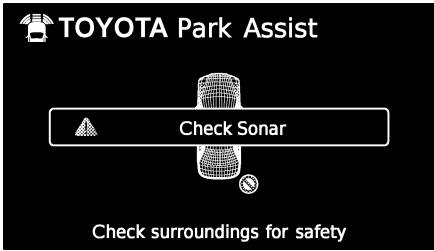
*: Although all areas with the sensors are marked in the illustrations, only the area with a malfunctioning sensor will be indicated by the mark in the actual condition.

4. Multi Display

- The items displayed by the multi-display are the location of the obstacle and the approximate distance between the vehicle and the obstacle, warning messages relating to sensor malfunction, sensor freezing, or presence of dirt on the sensor. When any warning messages are displayed, the master warning light illuminates and the clearance warning buzzer sounds.
- The distance is displayed by the number of lines and color based on the actual distance. The display usually illuminates in yellow. When the distance is short, it illuminates in red.



► Warning Message ◀



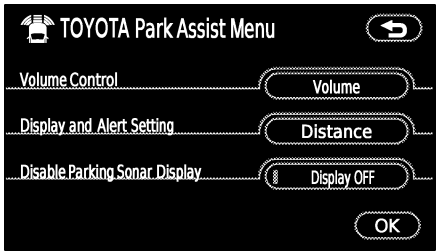
Ultrasonic Sensor Malfunction Warning



Ultrasonic Sensor Freeze/Contamination Warning

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- The display timing and detection distance on the multi-display can be customized.



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